



BISWAJEET KUMAR

Jamshedpur, Jharkhand, India

📞 7004602210 | ✉ biswajeetk497@gmail.com

🔗 GitHub: <https://github.com/Biswajeet111>

🔗 Portfolio: <https://biswajeet111.github.io/portfolio>

🔗 LinkedIn: <https://www.linkedin.com/in/biswajeet-kumar-a70043362>

Professional Summary

Computer Science Engineering student with hands-on experience in **full-stack development, Machine Learning, and AI-powered application development**. Skilled in building scalable web applications using **Python, FastAPI, React, and MongoDB**, with practical experience deploying production-ready systems on cloud platforms.

Developed multiple real-world projects including **AI-based study planning and financial analytics systems** integrating ML models and LLM-based insights. Passionate about solving real-world problems using intelligent systems and seeking internship opportunities in **Software Development, Full-Stack Development, or AI Engineering**.

Technical Skills

Programming Languages: C, C++, Java, Python, Dart, HTML, CSS, JavaScript, TypeScript

Frameworks & Tools: Flutter, Node.js, Android Studio, GitHub, Cisco Packet Tracer, FastAPI, React.js, Next.js, Tailwind CSS, Git, Vercel, Render, Postman

Databases: MySQL, Firebase, PostgreSQL, SQLAlchemy, MongoDB Atlas

Development Areas: Web Development, Mobile App Development, REST APIs, UI/UX Basics, Full-Stack Development, API Integration, Backend Development, Cloud Deployment

Soft Skills: Problem Solving, Team Collaboration, Leadership, Adaptability, Attention to Detail, Time Management, Communication, Critical Thinking

AI & Machine Learning: Machine Learning, Scikit-Learn, Model Deployment, Data Preprocessing, Feature Engineering, LLM Integration, Predictive Modeling

Internship Experience

Web Development Intern – Tata Steel

Jamshedpur | Duration: 1 Month

- Developed and maintained 5+ responsive web modules, improving UI consistency
- Used GitHub for version control and collaborative development
- Gained exposure to real-world software development workflows
- Improved code quality and UI consistency through iterative testing and feedback

Projects

ResQ – Emergency Medical Assistance Mobile Application

Diploma Final Year Project | Android Studio, Firebase, Google Maps API

- Designed and developed a multi-feature emergency response application
- Implemented secure user authentication and profile management

- Integrated location services to identify nearby hospitals using Maps API
- Built an AI-based symptom input module for preliminary medical guidance
- Enabled one-tap emergency calling and doctor contact access
- Focused on reducing emergency response time through automation

SchedSim – Intelligent CPU Scheduling Simulator

Operating Systems Project | Flutter, Dart

- Developed a **cross-platform CPU scheduling simulator**
- Implemented algorithms: **FCFS, SJF, Round Robin, Priority, SRTF, Multilevel Queue**
- Visualized scheduling using **Gantt charts and performance metrics**
- Compared **preemptive and non-preemptive algorithms** for efficiency analysis
- Enhanced learning experience through **interactive UI design**

AI Smart Expense & Payroll Management System

Full-Stack Financial Analytics Platform | React, FastAPI, PostgreSQL, SQLAlchemy, Vercel, Render

- Designed and developed an AI-powered financial management system for startups to automate payroll and track operational expenses.
- Implemented employee salary management with dynamic payroll calculation (base salary, bonuses, deductions).
- Developed expense tracking system with anomaly detection to identify unusual spending patterns.
- Built financial health scoring and burn rate analysis dashboard for real-time business insights.
- Integrated cash flow prediction logic to estimate future financial stability.
- Implemented REST APIs using FastAPI and managed relational data using SQLAlchemy ORM.
- Designed a professional SaaS-style dashboard UI with interactive charts and KPI indicators.
- Enabled downloadable PDF financial reports for payroll and expense summaries.
- Deployed production-ready full-stack application using Vercel (Frontend) and Render (Backend).

StudyForge AI — AI-Powered Personalized Study Planner (ML + LLM Based System)

Tech Stack: FastAPI, Python, MongoDB, Next.js, Scikit-Learn, LLM APIs

- Developed an AI-powered personalized study planner that generates optimized schedules using Machine Learning predictions
- Built REST APIs using FastAPI to manage study plans, progress tracking, and analytics
- Integrated LLM-based insights to provide personalized learning recommendations
- Implemented performance prediction using Scikit-Learn models
- Designed an interactive dashboard with progress tracking, XP gamification, and risk analysis
- Deployed the system on cloud infrastructure with external ML model hosting

EDUCATION

B. Tech in Computer Science Engineering

Lovely Professional University | Expected 2028 | 7.84 TGPA

Diploma in Computer Science Engineering
Government Polytechnic, Dhanbad | 2025 | **76%**

Senior Secondary (12th)
NIOS, Dhanbad | 2025 | **83.4%**

Secondary (10th)
CBSE | 2022 | **82.4%**

Languages Known

English: Professional

Hindi: Fluent

Hobbies & Interests

- Online Gaming (Strategic Thinking, Problem Solving)
- Technology & Software Innovations